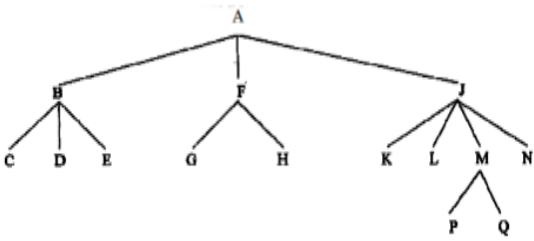


DSA:: Mock In Sem (2024-25) Sem2 : Question Bank

Q. No.	BL	CO	Questions	Unit No	Marks Weightage																		
1	3	1	Create a Hash table and resolve collisions using linear probing with and without replacement : 9, 45, 13, 59, 12, 75, 88, 11, 105, 46 Hash table size = 10 and Hash function = key mod 10	1	8																		
2	1	1	what is the use of hash Table	1	5																		
3	1	1	Write a note on characteristics of good hash function.	1	5																		
4	1	1	Explain rehashing concept with example.	1	5																		
5	1	1	Define the terms:collision,probe,synonym,overflow, perfect hash function, Load density	1	5																		
6	2	1	what is hash fuction?Enlist characteristics of good hash fuction.Explain modulo Division and folding method	1	5																		
7	1	1	what is collision?what are the different collision resolution Technique.	1	5																		
8	3	1	Given the Inputs : {4371, 1323, 6173, 4199, 4344, 9679, 1989} Hash table size =10 and Hash function = key mod 10. show the results for the following: i)Open addressing hash table using Linear Probing ii)Open addressing hash table using Quadratic Probing iii)open addressing hash table with second hash function h2 (X)=7-(x mod 7)	1	5																		
9	3	1	Explain Extendible hashing with suitable example.	1	5																		
10	1	1	Write a short note on separate chaining	1	5																		
			UNIT-II																				
11	3	2	Given the following traversals build a binary tree from them : (i) In-order : 15, 22, 25, 30, 33, 40, 44, 50, 60, 75, 80, 90 (ii) Pre-order : 50, 25, 22, 15, 40, 30, 33, 44, 75, 60, 90, 80.	2	6																		
12	3	2	Create a Huffman’s tree for the given data set and find the corresponding Huffman’s codes : <table><thead><tr><th>Data</th><th>Weight</th></tr></thead><tbody><tr><td>A</td><td>10</td></tr><tr><td>B</td><td>3</td></tr><tr><td>C</td><td>4</td></tr><tr><td>D</td><td>15</td></tr><tr><td>E</td><td>2</td></tr><tr><td>F</td><td>4</td></tr><tr><td>G</td><td>2</td></tr><tr><td>H</td><td>3</td></tr></tbody></table>	Data	Weight	A	10	B	3	C	4	D	15	E	2	F	4	G	2	H	3	2	8
Data	Weight																						
A	10																						
B	3																						
C	4																						
D	15																						
E	2																						
F	4																						
G	2																						
H	3																						

13	2	2	What is the necessity of converting a tree into binary tree ? Convert the following tree into a binary tree and list down the steps for the same. 	2	6
14	1	2	Define Tree and following terms with respect to tree with example 1. Root node 2. Siblings 3. Skewed Binary Tree 4. Complete Binary Tree	2	6
15	1	2	Explain Traversing of binary tree?	2	6
16	2	2	Explain Representation methods of binary tree in memory	2	6
17	2	2	Explain threaded binary tree and its representation in link list	2	6
18	2	2	Explain deletion of node in threaded binary tree	2	6
19	2	2	Write a recursive C++ function for recursive Preorder	2	4
20	2	2	Write an algorithm & pseudo code for binary tree recursive In-Order traversals	2	4

Subject Incharge
A. P. Shinde